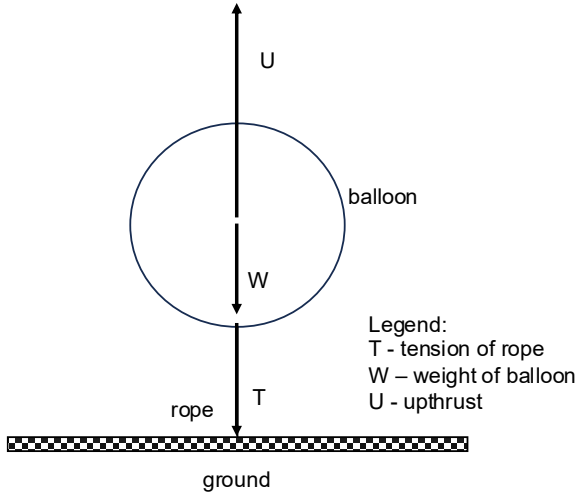


Section A

1a	Upthrust is the <u>resultant vertical upward force</u> exerted by a fluid on an object immersed in the fluid as the <u>pressure at the bottom surface is greater than that at the top</u> (or due to the <u>pressure difference between the top and bottom surfaces</u> of the object)	B1
bi	 Legend: T - tension of rope W - weight of balloon U - upthrust <u>Marking Points:</u> • correct direction with labeling (accept usual symbols U, W, T) • correct length/magnitude of U, W and T (length U about equal to W and T) and correct point of action of forces	
bii	$U = \rho Vg$ $= (1.3)\left(\frac{4}{3}\pi\left(\frac{1.5}{2}\right)^3\right)(9.81) = 22.54N$ $U = T + mg$ $T = U - mg$ $= 22.54 - (0.3)(9.81)$	C1 C1
	T = 19.6 N	A1
c	Net force upwards as <u>upthrust is more than weight</u> , so balloon rises	B1
	As <u>altitude increases, air pressure/density decreases, causing upthrust to decrease</u>	B1
	When (magnitude of) <u>upthrust is equal to weight</u> , balloon reaches its equilibrium height	B1
	Total:	9

